Paleomagnetic data from Early Cretaceous volcanic rocks of West Liaoning: Evidence for intracontinental rotation

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Abstract **Detailed rock magnetic studies of 55 lavas from Yixian and Fuxin area, West Liaoning, show the primary carriers of remanence to be pseudo-single domain titanomagnetite. K/Ar dating indicates that the volcanic sequence spans 93 to 133 Ma. Stepwise thermal demagnetization successfully isolated well-defined characteristic magnetization (ChRM) in all lavas thermal-treated above 250℃. The mean paleodirections are D/I=5.9□/58.8□(α_{95} = 2.9□) and D/I =179.2□/–59.9□(α_{95} = 5.2□) for 27 normally magnetized flows and 28 reversibly magnetized flows, respectively. It indicates that since the Early Cretaceous there is no significant horizontal movement and rotation between the Yixian-Fuxin area and Eurasia. However, Korea Peninsula may have undergone a clockwise rotation of 33.9□relative to the Yixian-Fuxin area during the Cretaceous. On the basis of characteristics of hotspot origins (core-mantle boundary or upper mantle), the clockwise rotation of Korea Peninsula relative to Eurasia is assumed to be mainly caused by an extensional force in the crust of eastern China, which was corresponding to intensive surface volcanic activities in this area.**

Keywords: **Cretaceous**, **paleomagnetism, geodynamics.**

Eastern China had been undergone tectonic structure changes from an E-W pattern in Paleozoic and early Mesozoic to a NNE pattern in late Mesozoic, and transition from a compression system to an expansion system. Meanwhile, in this region also occurred both deep continental subduction and the crust shortening[1]. Numerous metal deposits, oil and gas resources were accumulated, which is supposed to be closely related with the intra-continental tectonic processes since the Mesozoic. Therefore, investigations on continental tectonic history of eastern China are obviously very important for the study of continental dynamics. This note aims to study the kinematics characteristics of intracontinental blocks in eastern China since the Mesozoic by means of paleomagnetism.

1 Geological settings and sample collection

Eastern China underwent frequent intensive volcanic eruptions resulting from a complex influence of the western Pacific plate and deep Earth's processes during the middle to late Mesozoic. It forms a part of the Pacific-rim volcanic belt. In the Yixian and Fuxin region, it consists of Cretaceous basalts, andesites, trachytes, rhyolites, etc. The famous Jehol Fauna, including numerous early birds, flowers and mammals[2,3], which attracts great attention of paleobiologists and provides good materials for studying the biological massive extinctions, generations, the deep Earth's processes and ancient environments[4—7].

Total 410 orientated basaltic cores were collected from 55 lavas at the Zhuanchengzi section in Yixian, the Sihetun section at Beipiao, the Hulahada and Wuhuanchi in Fuxin, using a portable gasoline drill. All cores were orientated *in situ* using a magnetic compass and/or a sun compass. Each core was cut into 3—5 samples (1 cm in height) in laboratory for rock-magnetic and paleomagnetic measurements. In view of the stratigraphical sequence relative to the fossil-bearing lake sediments, basaltic lavas of the Yixian Formation at the Sihetun section consist of lavas overlying (#Sht1-2), intruded (#Sht3) and underlying ones (#Sht5-20); basaltic and andesitic intervals of the Yixian Formation at the Zhuanchengzi section (#ZCZ1-20). The Hulahada section and the Wuhuanchi section in Fuxin are parts of the Tuhulu Formation and the Dalinghe Formation, respectively.

2 K-Ar dating

K-Ar dating was conducted in the Geochronology Laboratory of the Institute of Geology, China Seismological Bureau. Fresh volcanic rocks were crushed and sieved to a size range between 0.20—0.28 mm. Mineral crystals, such as olivine and plagioclase, which may contain excess argon, were removed. The sieved samples were washed with water, ethanol and acetone, and then baked at low temperature.

Argon was analyzed using the isotope dilution technique with a 99.98% pure ^{38}Ar spike, on an MM1200 mass spectrometer connected to a purification and extraction system. The spike was calibrated and corrected with standard argon minerals: B4M and ZBH. K content was measured by a flame photometer (Model HG-3).

The samples were enclosed in a "Christmas tree"-shaped holder and heated to about 200℃ for more than 12 h, then separated and placed in individual molybdenum crucibles surrounded by a titanium crucible. Argon was extracted using an electron bombardment furnace with two vacuum systems: the inner vacuum system, including the Christmas tree, was connected to the purification system and mass spectrometer. The outer vacuum system was evacuated with a diffusion pump and used for electron bombardment heating. The samples were then

heated to 1320℃. Released gases were purified, first by a cold trap to remove CO_2 and H_2O, then by a titanium sponge to separate other active gases, and finally by Zr-Al getters. The purified noble gases were introduced into the mass spectrometer to measure the argon isotopes; K content was analyzed by an HG-3 flame photometer with Li inner standard. The K-Ar analytical data and resulting age determinations from the Yixian and Fuxin are presented in table 1.

The ages for those lavas (table 1) are between about 133 and 90 Ma (approximately 43 Ma). It indicates that the lake, in which the early birds and mammals were well-preserved, existed between 126—124 Ma (table 1), suggesting the massive extinction in a short interval. It also implies that the massive extinction might be associated with the local intensive volcanic eruptions and consequent local abrupt climatic changes.

3 Rock-magnetic properties and paleomagnetic results

(ⅰ) Rock-magnetic investigation. In order to determine the magnetic mineral characteristics of grain size and species in the studied lavas, detailed rock-magnetic investigations were conducted on at least two representative samples from each lava flow. All experiments were performed at the Paleomagnetism Laboratory of the Institute of Geology and Geophysics (IGG-CAS) in Beijing. Magnetic hysteresis loop determinations were conducted on small chips of samples using a Micromag 2900 alternating gradient force magnetometer (AGFM). After removal of a small paramagnetic contribution (given by the slope at high fields) values of H_{cr}, H_r, M_s, and M_{rs} were obtained. The ratios of M_{rs}/M_s and H_c/H_{cr} show the primary carriers of remanence to be pseudo-single domain (fig. 1). Isothermal remanent magnetization (IRM) acquisition curves revealed that the studied samples were nearly saturated by a field of 300 mT and values of the coercivity of remanence (H_{cr}) ranged from 15 to 40 mT, indicating magnetite as a main magnetic carrier (fig. 1).

Thermomagnetic (*J-T*) curves of powdered samples from each flow were measured in a steady field of 800 mT using a Magnetic Measurement Variable Field Translation Balance (MMVFTB). Based on the results, two groups of samples are identified (fig. 2). Titanium-poor magnetites, with a Curie temperature between 530—580℃ and a good reversibility of heating and cooling curves, suggesting that no distinct magnetic mineralogical alteration occurs during heating, were presented mainly in the Yixian Formation (fig. 2). Titanium-rich magnetites, with a Curie temperature below 300℃ were dominated in the basalts in the upper part of the Wuhuanchi section (Dalinghe Formation). The significant irreversibility of heating and cooling curves suggests that titanomagnetite/titanomaghemite may be transformed into hematite or maghemite during heating in these samples, whose cooling curves were much lower than heating curves (fig. 2).

(ⅱ) Paleodirectional results. Stepwise thermal demagnetization up to 585℃ in 13 steps of 25—50℃ was carried out using a Magnetic Measurements furnace MMTD60 (residual field <10 nT). Remanent measurements were made on a 2G cryogenic magnetometer

Table 1 K-Ar ages determined from Sihetun, Zhuanchengzi[7], Hulahada and Jianguo sections

Sample	K(%)	Weight/g	^{40}Ar (%)	^{40}Ar (10^{-10}mol/g)	^{38}Ar (10^{-13}mol)	^{40}Ar/^{38}Ar	^{38}Ar/^{36}Ar	Age/Ma
Sihetun								
SHT-1	2.17	43.40	96.75	4.837	1.906	113.84	74.34	124.16±2.4
SHT-3	2.185	51.20	96.44	4.881	1.908	135.83	56.03	124.42±2.4
SHT-11	2.10	38.75	98.42	4.710	1.904	97.41	180.00	124.91±2.4
SHT-13	1.57	57.75	91.77	3.557	1.9025	117.70	28.40	126.14±2.6
SHT-14	1.56	34.15	95.20	3.626	1.901	68.46	86.06	129.29±2.5
SHT-15	1.51	40.30	93.74	3.507	1.890	79.77	56.35	129.17±2.6
SHT-16	1.46	54.05	97.10	3.502	1.896	102.86	92.65	133.28±2.6
SHT-18	1.50	39.15	96.81	3.607	1.8855	77.38	113.72	133.59±2.6
Zhuanchengzi								
ZCZ-1	1.85	33.00	95.66	4.011	1.945	71.13	91.40	120.87±2.3
ZCZ-4	2.52	42.50	97.52	5.531	1.948	123.72	88.92	122.31±2.3
ZCZ-7	2.26	38.55	97.35	4.882	1.943	99.47	105.18	120.44±2.3
ZCZ-11	2.14	39.45	97.81	4.644	1.940	96.58	131.17	120.99±2.3
ZCZ-13	2.25	41.70	98.10	4.859	1.937	106.66	135.90	120.42±2.3
ZCZ17	2.36	38.75	95.24	5.100	1.934	107.32	54.07	120.51±2.3
ZCZ20	2.35	31.20	98.55	5.100	1.928	83.74	230.52	120.98±2.4
Hulahada and Jianguo								
JG-1	1.72	29.75	89.52	2.854	1.8603	50.999	53.637	93.32±1.96
JG-9	3.09	39.40	98.23	6.060	1.8584	130.76	117.12	109.7±1.0
FH-8	2.04	34.25	93.01	4.352	1.8575	86.268	46.469	119.0±2.4

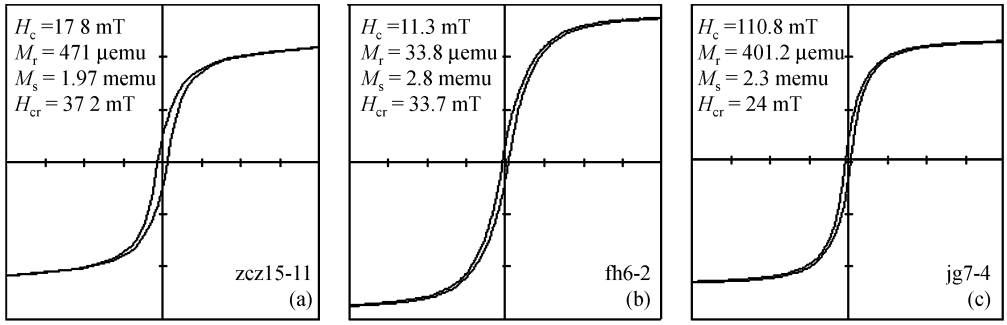


Fig. 1. Representative hysteresis loops. Samples zcz15-11, fh6-2 and jg7-4 were taken from the Zhuanchengzi, Hulahada and Wuhuanchi sections, respectively.

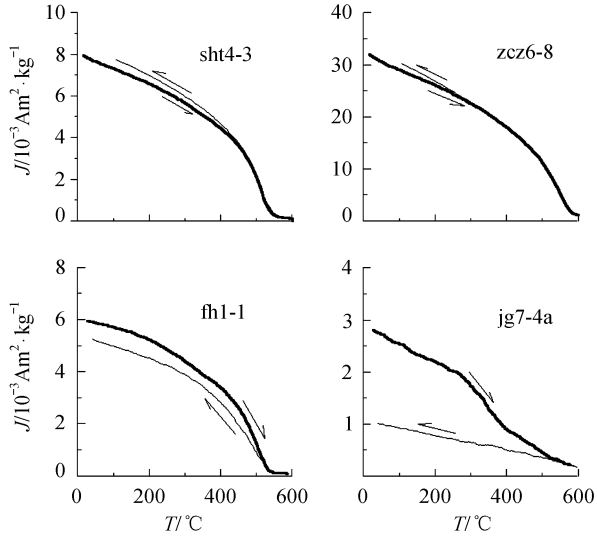


Fig. 2. Representative results of thermomagnetic curves (*J*-*T*), dark (light) curves stand for heating (cooling). Samples sht4-3, zcz6-8, fh1-1 and jg7-4a were taken from Sihetun, Zhuanchengzi, Hulahada and Wuhuanchi sections, respectively.

situated in a field free space (<300 nT). The majority of the samples are characterized by a stable characteristic remanent magnetization (ChRM) component isolated after demagnetization over 250℃. The stable component towards the origin of the orthogonal projection, indicating this component stands for the primary remanence magnetization (fig. 3). No reliable component can be isolated when the heating temperature is higher than 585℃.

The ChRM direction from each sample was determined using a least-squares method. Individual sample directions were averaged for each flow and statistical parameters calculated assuming a Fisherian distribution. After then mean direction for each section was obtained by averaging individual flow direction (table 2). As shown in table 2, normal and reversed directions were independently averaged, and passed the reversal test (f = 2.09 < F (2,14)=4.74).

4 Discussion and conclusions

A compilation of available palaeomagnetic data indicates that the North China block (NCB), South China block (SCB) and the Korea block had been completely sutured as a whole by the late Jurassic[8—11]. Whether there are relative intra-plate rotations since then can be revealed by means of paleomagnetism. To further this study, published palaeomagnetic data for NCB and Korea blocks since the Cretaceous (table 3) were compiled according to the present international criteria for reliability[12, 13]. For the NCB block and the Korea block, table 3 shows that no significant difference exists between their Cretaceous paleolatitudes and present latitudes. Furthermore, results of the Cretaceous basalts (for age spans 93—113 Ma see table 2) from Yixian and Fuxin indicate that West Liaoning has neither significant latitude-directional movement nor tectonic rotations relative to the main Eurasia continent since the Cretaceous (tables 2 and 3). The same features are observed for Inner Mongolia from paleomagnetitic study on the tuff of Pingzhuang. The Korea block has no horizontal movement and rotation relative to Eurasia since the Tertiary[10]. However, it is noted that there was a significant clockwise rotation of 33.9° relative to the Eurasia during the Cretaceous time[10] (North and South Korea have the same movement characteristic, see table 3).

Then what an internal force drives Korea to rotate clockwise during the Cretaceous time? West Liaoning and its adjacent areas and the Korea Peninsula may provide some useful clues. Strong tectonic movements and intensive volcanic activities were then their main characteristics of this region. Especially, in the early Cretaceous massive basaltic magma erupted in downfaulted basins[16]. Besides, the difference of the ratio of Ta/Hf and Th/Hf (table 4 and fig. 4) suggests that those basalts were formed in the expansional rift tectonic environment[17]. Therefore, we assume that the clockwise rotation of 33.9° of the Korea Peninsula was a result of the continental crust extension of eastern China, which was manifested by massive surface volcanic eruptions. Controlled by the

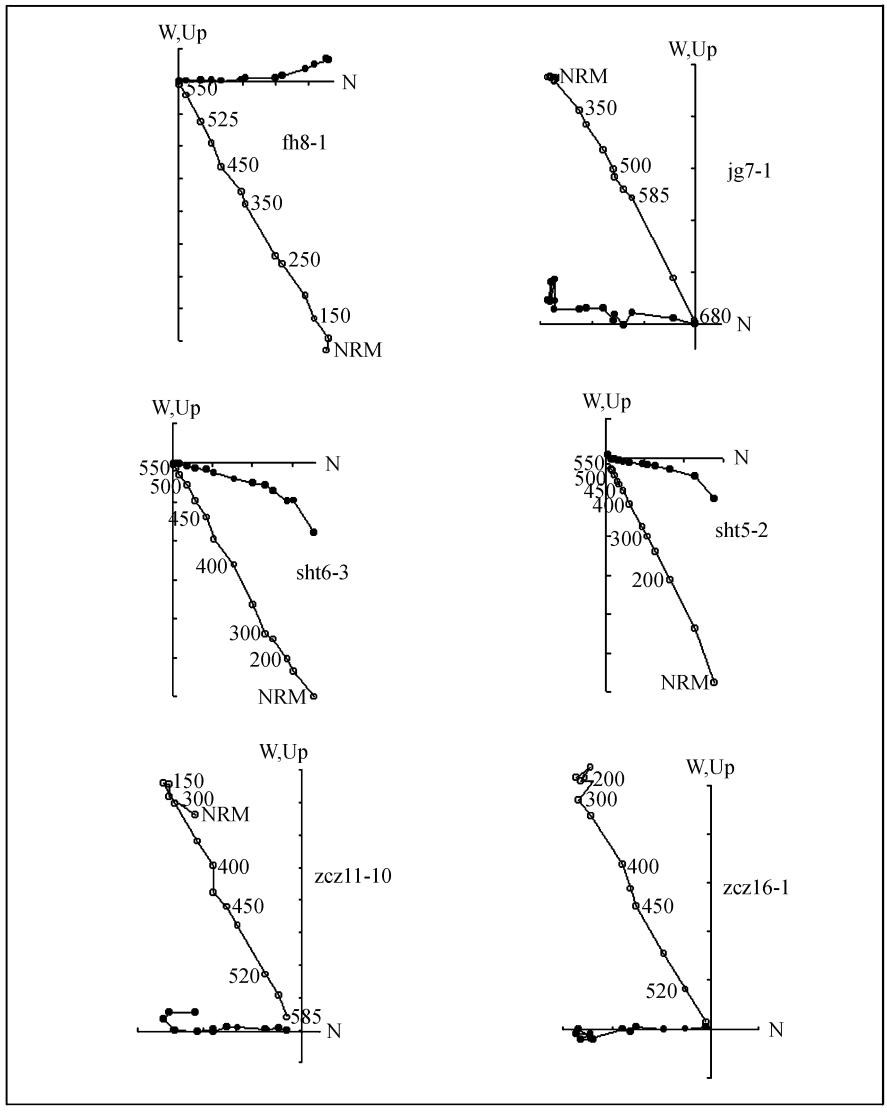


Fig. 3. Orthogonal projections of representative progressive thermal (℃) demagnetization. The solid (open) symbols refer to the horizontal (vertical) plane. W, Up and N stand for west, up and north, respectively. Samples marked with fh, jg, sht and zcz were taken from Hulahada, Wuhuanchi, Sihetun and Zhuanchengzi, respectively.

Table 2 Paleomagnetic results

Section	Lava No.	*n*/*N*	*D*/(°)	*I*/(°)	*PLA*/ (°)	*PLO*/ (°)	α_{95}/(°)	Age/Ma ($\pm 2\sigma$)
JG-U	5	31/35	6	59.6	85.1	232.4	3.9	93.22±1.96
JG-L	5	29/33	186.4	–58.2	–84.1	64.9	4.9	109.7±1.0
FL	7	42/46	7.9	59.5	83.8	222.5	3.6	119.0±2.40
ZCZ	20	136/140	174.9	–60.1	–86.1	205.9	2.3	120.93±0.88
SHT1-3	3	22/25	175.9	–60	–86.8	199.1	6.4	124.29±1.69
SHT5-11	7	52/58	4.9	59.3	86	231.1	3.6	124.91±2.38
SHT12-13	2	11/11	14.3	56.1	77.9	230.1	8.2	126.14±2.55
SHT14-15	2	18/19	–4.5	59.5	86.4	11.6	13.5	129.23±1.79
SHT16-19	4	30/33	6.2	58.2	84.5	238.5	5.5	133.43±1.81
N-P	6-site		5.9	58.8	85.1	233.4	2.9	
R-P	3-site		179.2	–59.5	–88.6	146.3	5.2	
N&R	2-P		2.6	59.2	87.5	249.2		

n/*N*, number of samples used in the calculation/total number demagnetized; *D*/*I*, declination/inclination; *PLA*/*PLO*, VGP latitude/longitude; α_{95}, radius of circle of 95% confidence about the direction. JG-U/JG-L, upper/lower part of the Jianguo section; FL, Hulahada section; ZCZ, Zhuanchengzi section; SHT, Sihetun section; N-P/R-P, normal/reversed polarity, respectively.

Table 3 Published paleomagnetic poles for the North China block and Korea Peninsula

Site	S_{lat}/S_{long}	D/(°)	I/(°)	PLA/(°)	PLO/(°)	A_{95}/(°)	Age/Ma
Liaoxi	41.6/120.7	2.6	59.2	87.5	249.2		133.43—93.22
Pingzhuang	42/119.2	6.8	56.6	82.9	249.5	5.7	K1[14]
N Kor.	37/128	37.5	61.3	60.9	195.5	8.2	K[15]
S Kor.	35.9/128.6	36.5	59.4	61.2	199.5	6.6	Aptian[10]

Table 4 Ratios of Th/Hf and Ta/Hf for some volcanic rocks

Sample	Jg1	Jg2	Fh	Sht
Age	Up-K_{1-2}	Low-K_{1-2}	K_1	K_1
Rock	basalt	basaltic-andesite	basalt	basalt
Th/Hf	1.06	0.77	1.01	1.15
Ta/Hf	0.73	0.19	0.19	0.13

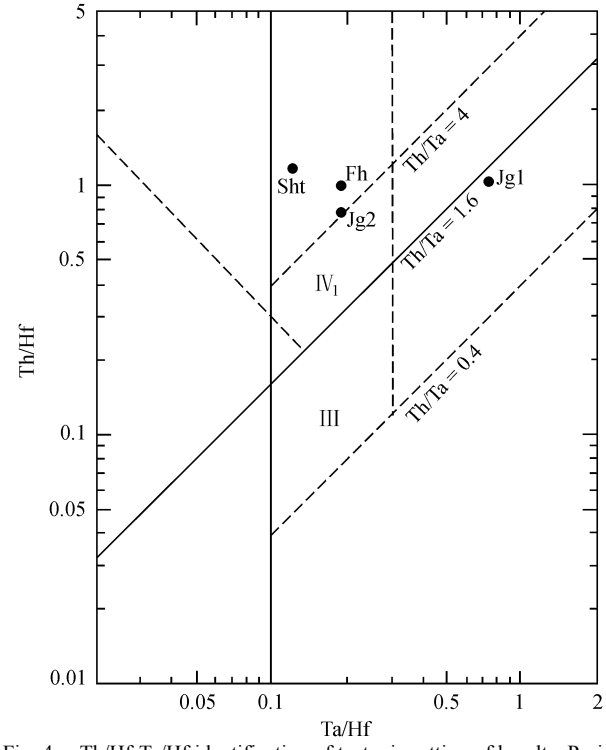


Fig. 4. Th/Hf-Ta/Hf identification of tectonic setting of basalts. Projection points Sht, Fh, Jg are for samples from Sihetun, Hulahada and Jianguo respectively.

main Erasia, West Liaoning and Pingzhuang area of Inner Mongolia may have had no significant effect on the crust extensional process.

Although most volcanic activities occur at plate margins (such as mid-oceanic ridge and subduction zone), some basaltic eruptions may happen at intraplates. Morgon (1972) proposed that hotspots are formed as the lower mantle-flow upwelling onto the surface, often distributed along the extensional plate boundaries and areas with high long-wave geoid[18]. Recently, Zhao (2001) reported that the mantle plumes beneath hotspots in Hawaii, Iceland, south Pacific and East Africa originate from the core-mantle boundary (CMB); there are also some hotspots corresponding to small-scale mantle plumes that originate from the discontinuity boundaries at depths of 410 or 610 km, and, the western Pacific margin is a zone with a low velocity of seismic wave[19]. Furthermore, computer simulation and experimental studies show that the temperature and physical property differences in the mantle beneath the edge of thick continental lithosphere and rift can produce local mantle convections[20], which may produce massive continental volcanic eruptions[21—23]. Combined with above observations of seismology and experimental simulation, we assume that extensively-distributed basalts in West Liaoning may be associated with a small-scale mantle plume originated from the upper mantle. The mantle plume is generated by an upwelling of local heat flow from the discontinuity boundaries at depths of 410 and 610 km.

This study shows that West Liaoning area relative to Eurasia has no significant horizontal movement and rotation since the early Cretaceous, but the Korea Peninsula had a clockwise rotation of 33.9° relative to Eurasia in the early Cretaceous time; in view of geochemistry, those basalts were produced in the extensional rift geotectonic environment.

A full explanation for the clockwise rotation of the Korea Peninsula is beyond the scope of this investigation. Remaining questions are if this rotation was related with the deep Earth's extension process, e.g. a plume from the discontinuity boundaries at depths of 410 and 660 km or from the upper mantle, or with the Mesozoic shear-strike slip movements at the East Asia continental margin. The fact of a clockwise rotation of 33.9° is clearly contradicted to some geological assumptions that sinistral shear movements at this area in Mesozoic time. To solve the above questions, a combined study by different approaches must be done.

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